

movement (e.g., up days and down days) in order to determine if there is any bias for certain conditions or trading characteristics (e.g., one broker or algorithm performs better for high volatility stocks, another broker or algorithm performs better in favorable trending markets, etc.).

COMPARING ALGORITHMS

One of the biggest obstacles in comparing algorithmic performance is that each algorithm trades in a different manner, under a different set of market conditions. For example, a VWAP algorithm trades in a passive manner with lower cost and more risk compared to an arrival price algorithm which will trade in a more aggressive manner and have higher cost but lower risk. Which is better?

Consider the results from two different algorithms. Algorithm A has lower costs on average than algorithm B. Can we conclude that A is better than B? What if the average cost from A and B are the same but the standard deviation is lower for A than for B. Can we now conclude that A is better than B? Finally, what if A has a lower average cost and also a lower standard deviation? Can we finally conclude that A is better than B? The answer might surprise some readers. In all cases the answer is no. There is simply not enough information to conclude that A is a better performing algorithm than B even when it has a lower cost and lower standard deviation. We need to determine whether or not this is a statistical difference or due to chance.

One of the most fundamental goals of any statistical analysis is to determine if the differences in results are “true” differences in process or if they are likely only due to chance. To assist with the evaluation of algorithms we provide the following definition:

Performance from two algorithms is equivalent if the trading results are likely to have come from the same distribution of costs.

There are two ways we can go about comparing algorithms: paired observations and independent samples.

A paired observation approach is a controlled experiment where orders are split into equal pairs and executed using different algorithms over the same time periods. This is appropriate for algorithms that use static trading parameters such as VWAP and percentage of volume (POV). These are strategies that will not compete with one another during trading and are likely to use the exact same strategy throughout the day. For example, trading 1,000,000 shares using a single broker’s VWAP algorithm will